

How is AI changing the way workers perform their jobs and the skills they require?

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Key findings

- Artificial Intelligence (AI) is changing how workers perform their jobs and what skills they require.
- In occupations most exposed to AI (e.g. computer programmers, budget analysts and administrative assistants), management and business skills are the most demanded skills. 72% of vacancies posted in these occupations demand at least one management skill (e.g. budgeting and finance) and 67% at least one business skill (e.g. clerical skills). In addition, 58% demand at least one digital skill.
- However, the demand for management, business and digital skills has been falling in the workplaces most exposed to AI: a three percentage point decline in vacancies demanding such skills over the past decade.
- The magnitude of this decline is relatively small (equivalent to a workplace posting one less vacancy demanding at least one of these skills). However, as AI adoption increases, demand for these skills should be monitored closely.

Artificial Intelligence (AI) is changing how workers perform their jobs and what skills they require. As an example, take the case of an insurance company that adopts an AI tool that uses historical data to predict when a customer is likely to escalate a service issue. Prior to the adoption of the tool, sales agents conducted spot checks on a random selection of customer files. The work required both cognitive and financial skills to identify potential issues, but also social and communication skills to effectively reach out to customers. After the introduction of AI, agents may be given a priority list of account issues which the AI tool had already identified. The job of a sales agent changed to emphasise greater customer interaction with less time needed to analyse customer files (Milanez, 2023^[1]).

This policy brief summarises first estimates of the changing skill demand from AI exposure across 10 OECD countries over the past decade (Green, 2024^[2]).¹ The data come from Lightcast and contain the quasi-universe of online job vacancies complete with descriptions of the key skills and competencies required. The Lightcast data is combined with data on AI exposure from Felten, Raj and Seamans (2021^[2]) who use information on basic AI advances collected from the Electronic Frontier Foundation's AI measurement project to calculate a relative ranking of occupational exposure to AI based on the overlap between an occupation's tasks and the capabilities of AI. Finally, vacancies demanding AI skills have been removed from the analysis, so the focus is on workers who will be using AI in their jobs, but not actively developing or maintaining AI systems.

One-third of vacancies are in occupations highly exposed to AI

Across the 10 OECD countries studied, about one-third of vacancies are in occupations highly exposed to AI (defined as occupations with an exposure measure at least one standard deviation above the mean) ranging from 31% in Austria to 45% in the United Kingdom. Occupations highly exposed to AI tend to require higher than average education (OECD, 2023^[3]), and include secretaries and administrative assistants, accountants and financial analysts, software developers, managers and human resources professionals.

Management and business skills are the most demanded skills in occupations highly exposed to AI

Management and business skills are the most demanded skill groupings in occupations highly exposed to AI, with 72% and 67% of vacancies in these occupations demanding these skills in 2021-22 (end-year) (Figure 1). Management skills include human resource management and general project management as well as accounting and budgeting skills. Business skills include administrative and clerical skills and skills in customer service and sales. These skill groupings (as well as digital skills) are required in white-collar office work to do tasks that might be described as "routine" but are difficult to codify in a deterministic way. These are the types of tasks where current AI applications excel.

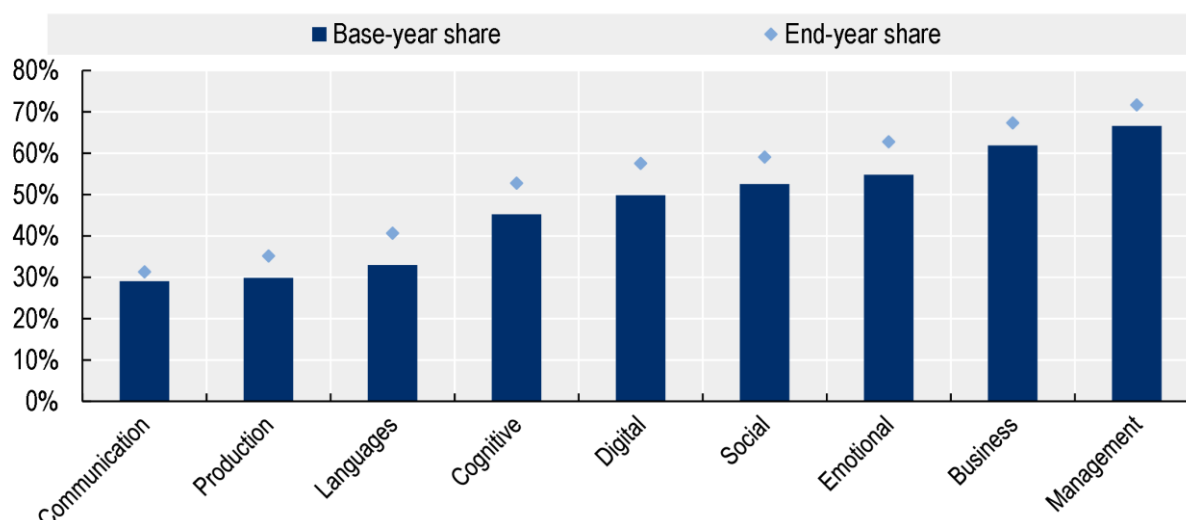
The most demanded skills in occupations highly exposed to AI have also seen some of the largest increases in demand over time. The demand for business and management skills have increased by about 8% over the period studied. Emotional, digital and social skills have experienced even greater increases in demand of approximately 15%. Finally, the demand for cognitive and language skills has also increased significantly in occupations highly exposed to AI. Cognitive skills include originality as well as reasoning and problem-solving skills, among others. Language skills are almost exclusively the ability to speak foreign languages, and in particular, English.

The demand for these skills has also increased in occupations less exposed to AI, which suggests that factors other than just AI may be driving these changing skills demands, such as the general trend towards increased digitalisation. For example, the general trend towards increased digitalisation and the shift from manufacturing to services means that skills most exposed to AI will be increasingly demanded in OECD economies, but this would have little to do with the proliferation of AI. Looking more closely at what happened to the demand for skills in workplaces more exposed to AI makes it easier to single out the effect of AI.

¹ The countries are Austria, Belgium, Canada, Czechia, France, Germany, the Netherlands, Sweden, the United Kingdom and the United States.

Figure 1. The most demanded skills in occupations highly exposed to AI have also seen some of the largest increases in demand

The most demanded skill groups in high exposure occupations



Note: Share is defined as the share of vacancies in high exposure occupation tercile demanding at least one of the skills from each skill grouping in each country. Datapoints are the unweighted average across countries. The countries included are the United States, Canada, the United Kingdom (English-speaking countries), and France, Germany, Belgium, Sweden, the Netherlands, Austria and Czechia (European countries). The base years for English-speaking countries are pooled 2012-13, and pooled 2018-19 for European countries. The end years are pooled 2021 and 2022.

Source: Green (2024^[4]), "Artificial intelligence and the changing demand for skills in the labour market", <https://doi.org/10.1787/88684e36-en>.

The demand for management, business and digital skills has fallen in workplaces most exposed to AI

Instead of looking at occupations, the attention shifts to workplaces where there is evidence of falling demand for some of the skills highlighted above. In workplaces more exposed to AI, the share of vacancies demanding management, business, digital and cognitive skills declined by over 3 percentage points over the past decade compared to workplaces that are less exposed (Figure 2). Focusing on workplaces that are observationally similar, in the same industry and of similar employment size, does a better job at identifying the causal effect of AI than focusing on occupations.

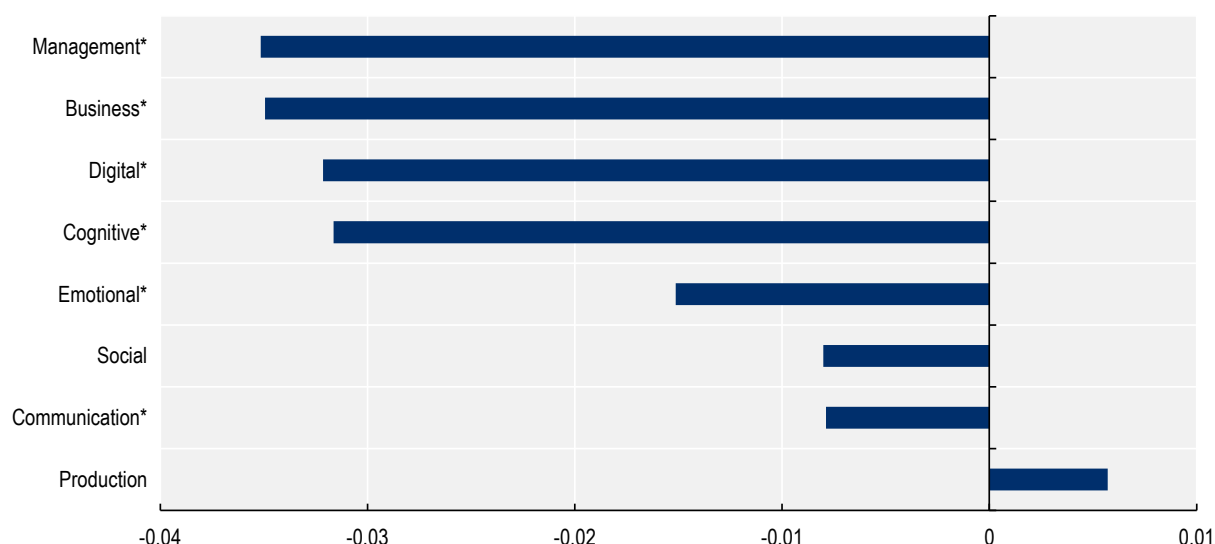
The magnitudes of these changes are relatively small, however, and are equivalent to each workplace posting around one less vacancy demanding at least one of these skills (the average workplace posts around 20 vacancies). However, should AI adoption continue to increase, one may see a bigger fall in demand for these skills going forward.

The analysis at the workplace level also allows one to capture changes in demand for skills that have little overlap with AI's capabilities. Returning to the example of the insurance company, the adoption of the AI tool should lead to greater productivity and cost savings for the firm. If the firm passes along these savings to customers, this may lead to greater demand for their insurance products. The firm may respond to this by hiring more sales agents, thereby increasing demand for social and communication skills. However, the firm may also need to hire more auto damage appraisers, for example. These are workers who appraise vehicle damage to determine repair costs for insurance companies. The work requires repair and

maintenance skills, which are more typical of auto mechanics and other blue-collar occupations with little AI exposure.

Figure 2. Demand for management, business and digital skills has fallen in workplaces more exposed to AI

Country average of regression coefficients for the percentage point change in demand for skill groupings from establishment-level AI exposure, by skill grouping



Note: Bars are unweighted cross-country average regression coefficients of establishment-level AI exposure in each country. They are interpreted as the percentage point change in the share of vacancies demanding at least one skill from the grouping between the base and end years for a one standard deviation increase in establishment-level AI exposure. Stars indicate the skill groupings where at least 7 of 10 countries in the average have i) regression coefficients that are the same sign as the cross-country average and ii) those regression coefficients are significant at the 95% confidence level.

Source: Green (2024^[4]), "Artificial intelligence and the changing demand for skills in the labour market", <https://doi.org/10.1787/88684e36-en>.

There is evidence that the demand for some of these blue-collar skills has increased in workplaces more exposed to AI. For example, compared to workplaces moderately exposed to AI, the demand for skills related to production has increased by 2 percentage points. The most frequently demanded skills in the production grouping are repair, cleaning, troubleshooting, and quality assurance and control. However, these increasing skill demands were more modest – a less than 1 percentage point increase for moving from a moderate to a highly exposed establishment – with substantial cross-country heterogeneity.

Adapting skills and training to the new realities of AI

As AI is progressively incorporated into the workplace, the demand for skills is changing as well. The advance of AI may result in declining demand for some skills that are disproportionately exposed to AI. Based on the results presented here, AI may also present opportunities for workers in blue collar occupations thanks to productivity spill-over effects. Skills and training policies will, therefore, need to adapt to the new realities of AI in the workplace. Recent evidence from the OECD finds that workers are significantly more likely to report positive outcomes of AI on their working conditions when they engage in training (Lane, Williams and Broecke, 2023^[5]). With consistent monitoring, an adapted skills and training offering and input from social partners, the AI era can bring rising productivity and living standards for all.

References

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Explore further

Read the related working paper:

Green, A. (2024), “Artificial intelligence and the changing demand for skills in the labour market”, *OECD Artificial Intelligence Papers*, No. 14, OECD Publishing, Paris, <https://doi.org/10.1787/88684e36-en>.

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Contact

Stefano SCARPETTA (✉ stefano.scarpetta@oecd.org)

Stijn BROECKE (✉ stijn.broecke@oecd.org)

This policy brief contributes to the OECD's Artificial Intelligence in Work, Innovation, Productivity and Skills (AI-WIPS) programme, which provides policymakers with new evidence and analysis to keep abreast of the fast-evolving changes in AI capabilities and diffusion and their implications for the world of work. The programme aims to help ensure that adoption of AI in the world of work is effective, beneficial to all, people-centred and accepted by the population at large. AI-WIPS is supported by the German Federal Ministry of Labour and Social Affairs (BMAS) and will complement the work of the German AI Observatory in the Ministry's Policy Lab Digital, Work & Society. For more information, visit <https://oecd.ai/work-innovation-productivity-skills> and <https://denkfabrik-bmas.de/>.



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